

Heterogeneous Satellite Mission Scheduling in Large-Scale Constellations With Transformer-Reptile MAPPO Approach

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Abstract—The emergence of sixth-generation (6G) communication technology motivates the evolution of satellite networks from simple single-function and independently operated systems toward hierarchical and intelligent cooperative satellite architectures. Such architectures introduce diverse businesses and heterogeneous missions, each imposing unique requirements on network resources and performance metrics. Traditional monolithic learning policies struggle in these heterogeneous environments due to negative transfer between mission-specific objectives. To address this challenge, we reframe satellite mission scheduling as a multi-task learning (MTL) problem, naturally capturing the shared structure across different missions. We propose Transformer-Reptile Multi-Agent Proximal Policy Optimization (TRMAPPO), integrating Transformer networks for inter-agent relationship modeling and Reptile for rapid policy specialization. Extensive experiments are conducted based on realistic satellite constellations and authentic ground-user datasets. Results demonstrate TRMAPPO significantly improves the success rate and reduces the latency of overall missions compared to state-of-the-art benchmarks, validating its effectiveness for hierarchical satellite cooperative systems.

Index Terms—Satellite networks, mission scheduling, multi-agent reinforcement learning (MARL), multi-task learning (MTL), proximal policy optimization (PPO).

I. INTRODUCTION

With the rapid development of 6G communication technology, satellite systems are undergoing a fundamental transformation from isolated, single-mission satellites to large-scale constellations [1]. This evolution is propelled by the burgeoning demands of future applications, such as auxiliary communications for disaster-affected areas [2] and higher-frequency remote sensing revisits [3]. Orchestrating these heterogeneous services introduces significant operational challenges, mainly due to the limited onboard resources, highly dynamic network topology, and the inherent difficulty in establishing reliable inter-satellite links. To address these challenges, satellite cooperation has emerged as a key enabling technology. By leveraging the spatiotemporal complementarity of idle resources and diverse coverage across the constellation,

cooperation can significantly enhance service delivery and overall system efficiency [4], making it a critical and hot topic of recent studies.

Recently, a large number of studies have focused on scheduling diverse missions in satellite cooperation system. For communication missions, Chai *et al.* [5] designed a fairness-based satellite relay algorithm to address the horizon communication problem of near-space vehicles. Zeng *et al.* [6] investigated relay selection in two-way hybrid terrestrial-satellite systems. They choose the node maximizing satellite received power to reduce outage probability. For satellite edge computing (SEC) missions, Lyapunov optimization [7] and alternating direction method of multipliers (ADMM) [8] have been used to optimize overall latency and energy consumption in computing offloading. In terms of remote sensing missions, Li *et al.* [9] developed a pre-trained reinforcement learning model (PTMB) based framework to address large-scale online remote sensing task scheduling problems. Long *et al.* [10] proposed a multi-satellite cooperative algorithm based on a co-evolution mechanism named MSCTP-CEM, to address complex earth observation problems.

Nonetheless, existing studies still treat the scheduling of each type of mission as an independent problem (e.g., communication relay, computation offloading, and sensing allocation). This overlooks the spatiotemporal coupling of resource occupancy across missions, which can lead to significant QoS degradation. For example, the continuous occupation of high-bandwidth Inter-Satellite Links (ISLs) by computation offloading of sea surface temperature sensing data will block real-time video calls for disaster-affected areas which requires strict QoS demands. Traditional conflict-resolution methods, like fixed resource partitioning, are ill-suited for the dynamic and resource-constrained satellite environment. To enable joint and efficient resource reuse among satellites, it is essential to develop a unified mission scheduling strategy instead of simply treating them as isolated problems.

Initial approaches relied on mathematical modeling to schedule a limited set of missions, such as communication

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and computation [11], [12]. These model-based methods lack scalability, making them difficult to generalize to broader satellite scenarios. To overcome these scalability issues, recent research has shifted towards learning-based methods like deep reinforcement learning (DRL) for their adaptability in complex and dynamic satellite environments [13]. However, generalizing a learned policy to diverse mission logics requires massive datasets for training, and the resulting performance can be unstable when encountering novel scenarios [14]. Motivated by this, we reframe the heterogeneous scheduling problem through the lens of multi-task learning (MTL). We propose a novel meta-learning-enhanced MARL framework, termed Transformer-Reptile Multi-Agent Proximal Policy Optimization (**TRMAPPO**), to learn a generalizable policy across diverse satellite missions. Main contributions are listed as:

- 1) We formulate the complex scheduling of heterogeneous satellite missions (e.g., communication, computation, and remote sensing) as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP). To enable effective policy generalization across these diverse tasks, we approach this problem through a multi-task learning paradigm. To the best of our knowledge, this work represents the first systematic effort to solve the dynamic joint mission scheduling problem in satellite networks from an MTL perspective.
- 2) To tackle the challenges of high environmental dynamism and constrained ISLs, we apply a novel Transformer-based MAPPO architecture. The self-attention mechanism empowers each agent to infer the broader system state and the intentions of other agents using only its local observations. This significantly enhances collaborative decision-making and adaptability without depending on frequent, explicit communication.
- 3) We integrate the Reptile meta-learning algorithm into the MARL framework to endow the scheduling policy with rapid adaptation and generalization capabilities for multi-task learning. Extensive simulations validate the superiority of TRMAPPO over five baseline methods. Our approach achieves a 13.02% improvement in mission completion ratio and an 8.84% reduction in average delay, demonstrating its remarkable effectiveness and robustness across diverse load distributions.

The remainder of this paper is organized as follows. Section II details the system model and problem formulation. Section III introduces the proposed TRMAPPO algorithm. Section IV validates the framework through comprehensive simulations, and Section V concludes the paper.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Network Model

As shown in Fig. 1, the hierarchical architecture consists of a ground segment $\mathcal{G}$ and three orbital segments: low earth orbit (LEO), sun-synchronous orbit (SSO), and geostationary orbit (GEO). The satellite set is described as $\mathcal{S} = \mathcal{S}_{\text{LEO}} \cup \mathcal{S}_{\text{SSO}} \cup \mathcal{S}_{\text{GEO}}$, where each subset corresponds to

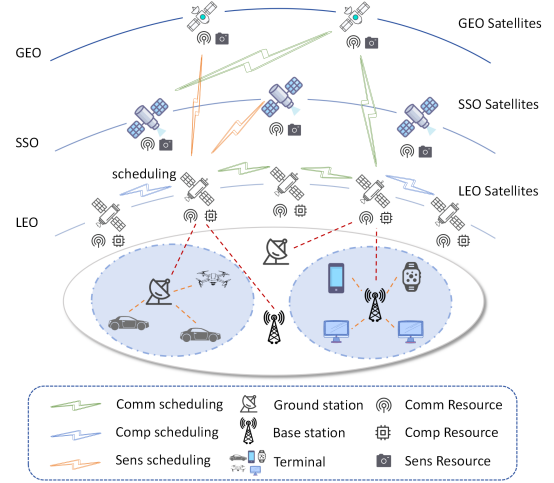


Fig. 1: Architecture of Hierarchical Intelligent Satellite Cooperation System.

nodes in a specific orbital layer, and the overall node set is $\mathcal{V} = \mathcal{S} \cup \mathcal{G}$. LEO satellites enable delay-sensitive services via high-throughput ISLs, SSO satellites support Earth observation and preprocessing, and GEO satellites provide high-capacity relay links for latency-tolerant missions, while ground stations aggregate and pre-process user requests. Position of node u is represented in the Earth-Centered Earth-Fixed (ECEF) coordinate system as $\mathbf{p}_u = (x_u, y_u, z_u)$.

B. Communication Model

The entire simulation process is divided into T discrete time slots, each representing a fixed-duration interval during which the network topology is assumed to remain constant. Within each time slot $t \in \{1, 2, \dots, T\}$, the communication connectivity between network nodes is represented by a binary line-of-sight (LoS) indicator $\ell_{u,v}(t) \in \{0, 1\}$ for each node pair $(u, v) \in \mathcal{V} \times \mathcal{V}$. A link exists ($\ell_{u,v}(t) = 1$) if and only if the straight line connecting nodes u and v is not obstructed by the Earth; otherwise, $\ell_{u,v}(t) = 0$.

The achievable data rate for an established link between nodes u and v is determined by the Shannon-Hartley channel capacity theorem, given by:

$$C_{u,v}(t) = B_u \log_2 \left(1 + \frac{p_u |h_{u,v}(t)|^2}{\sigma^2} \right), \quad (1)$$

where B_u and p_u denote the channel bandwidth and transmission power of satellite u , $|h_{u,v}(t)|^2$ represents the channel power gain that incorporates the Doppler effect, and σ^2 is the noise power.

C. Task Model

We consider three types of missions, namely *communication*, *computation*, and *sensing*. Each type consists of a mission set generated by terrestrial users, and we denote the overall mission set as $\mathcal{K} = \mathcal{K}^{\text{COMM}} \cup \mathcal{K}^{\text{COMP}} \cup \mathcal{K}^{\text{SENS}}$, where $\mathcal{K}^{\text{COMM}}$, $\mathcal{K}^{\text{COMP}}$, and $\mathcal{K}^{\text{SENS}}$ represent the set of communication, com-

putation, and sensing missions, respectively. Each mission $k \in \mathcal{K}$ is represented by:

$$\Theta_k = \{\tau_k, \alpha_k, D_k^{\max}, u_k^{\text{src}}, t_k^{\text{arr}}\}, \quad (2)$$

where $\tau_k \in \mathcal{T}_{\text{types}} = \{\text{COMM}, \text{COMP}, \text{SENS}\}$ denotes the mission type, α_k denotes the data size of mission k , $D_k^{\max}$ represents maximum delay tolerance, node u_k^{src} identifies mission originating node, and t_k^{arr} specifies arrival time. Additionally, sensing-oriented missions include auxiliary parameters such as target coordinates, observation duration, and sensing accuracy.

Considering a generic mission scheduling route $\mathcal{P}_k = (u_0, \dots, u_m)$, the end-to-end latency comprises transmission, propagation, queuing, and processing delays. Transmission and propagation delays are cumulative over each hop. The one-hop communication delay between nodes u_i and u_{i+1} at time t is:

$$L_{k,u_i,u_{i+1}}^{\text{comm}}(t) = \frac{\alpha_k}{C_{u_i,u_{i+1}}(t)} + \frac{d_{u_i,u_{i+1}}(t)}{c_0}, \quad (3)$$

where $d_{u_i,u_{i+1}}(t)$ is the instantaneous distance between communicating nodes. Let L_{k,u_i}^{proc} denote the processing delay at satellite u_i . We distinguish between computation and sensing missions, with respective resource allocation constraints. The queuing delay L_{k,u_i}^{queue} at a node n accounts for waiting times under a FIFO processing discipline, since the mission k arrive at node k at t_{k,u_i}^{arr} . Thus, the end-to-end latency for mission k is summarized by:

$$L_k = \sum_{i=0}^{m-1} L_{k,u_i,u_{i+1}}^{\text{comm}}(t_i) + \sum_{i=1}^m L_{k,u_i}^{\text{queue}} + L_{k,u_m}^{\text{proc}}. \quad (4)$$

D. Problem Formulation

The objective is to find an schedule policy that minimizes the weighted sum of average mission completion latency and maximizes the mission completion rate. This is a multi-objective optimization problem. Let $\mathbf{f} = \{f_{u,v}^k\}$ denote the scheduling decisions, where $f_{u,v}^k$ indicates that satellite v is selected to assist in completing mission k and $u = u_0^k$ denotes the source node of mission k . The problem is formulated as:

$$\min_{\mathbf{f}} \quad \omega_1 \left(\frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \mathcal{L}_k(\mathbf{f}) \right) - \omega_2 \left(\frac{|\mathcal{K}_{\text{succ}}|}{|\mathcal{K}|} \right) \quad (5a)$$

$$\text{s.t.} \quad \sum_{v \in \mathcal{S}_{\text{LEO}}} f_{u,v}^k = 1, \quad \forall k \in \mathcal{K} \text{ with } \tau_k = \text{COMP}, u = u_0^k \quad (5b)$$

$$\sum_{v \in \mathcal{S}_{\text{SSO}} \cup \mathcal{S}_{\text{GEO}}} f_{u,v}^k = 1, \quad \forall k \in \mathcal{K} \text{ with } \tau_k = \text{SENS}, u = u_0^k \quad (5c)$$

$$\mathcal{L}_k(\mathbf{f}) \leq D_k^{\max}, \quad \forall k \in \mathcal{K}_{\text{succ}} \quad (5d)$$

$$f_{u,v}^k \leq \ell_{u,v}(t), \quad \forall k \in \mathcal{K}, \forall v \in \mathcal{V}, u = u_0^k \quad (5e)$$

$$f_{u,v}^k \in \{0, 1\}, \quad \forall k \in \mathcal{K}, \forall v \in \mathcal{V}, u = u_0^k \quad (5f)$$

Here, $\mathcal{L}_k$ is the total latency of mission k , $\mathcal{K}_{\text{succ}}$ denotes the set of completed missions, and ω_1, ω_2 are weight coefficients. Constraint (5b) ensures that each computation mission is assigned to one LEO satellite from the set $\mathcal{S}_{\text{LEO}}$. Constraint (5c)

assigns each sensing mission to an appropriate sensing satellite, which may belong to $\mathcal{S}_{\text{SSO}}$ or $\mathcal{S}_{\text{GEO}}$. Constraint (5d) ensures timely mission completion. Constraint (5e) restricts transmission to available communication links, and (5f) define the domains of decision variables.

III. TRMAPPO: A MULTI-TASK MARL FRAMEWORK FOR SATELLITE MISSION SCHEDULING

A. DEC-POMDP Formulation

The mission scheduling problem is formulated as a Decentralized Partially Observable Markov Decision Process (DEC-POMDP), which is formally defined by the tuple $\langle \mathcal{I}, \mathcal{S}, \{\mathcal{A}_i\}_{i \in \mathcal{I}}, \{\Omega_i\}_{i \in \mathcal{I}}, \mathcal{P}, \mathcal{R} \rangle$, where $\mathcal{I}$ is the finite set of N satellites.

1) *Observation Space* (Ω_i): Each agent i receives a partial observation $o_{i,t} \in \Omega_i$ at time step t , which is structured as a tuple $o_{i,t} = (\mathbf{T}_t, \mathbf{P}_t, \mathbf{R}_t)$. Here, $\mathbf{T}_t$ includes mission characteristics such as type, data size, and required computation workload; $\mathbf{P}_t$ describes the spatial and dynamic state of agent i and its one-hop neighbors $\mathcal{N}_i(t)$; and $\mathbf{R}_t$ contains local resource features, including the idle ratio of communication beams and available CPU cores.

2) *State Space* ($\mathcal{S}$): The overall state space $\mathcal{S}$ is defined as the set of all possible observations that agents may receive, i.e., $\mathcal{S} = \bigcup_i \Omega_i$.

3) *Action Space* ($\mathcal{A}_i$): The action $a_{i,t} \in \mathcal{A}_i$ of agent i determines how to handle the mission currently at the head of its local queue. The action space is discrete and dynamic. Let $\mathcal{N}_i(t)$ be the one-hop neighbor set of agent i at time t , then the action set is given by:

$$\mathcal{A}_i(t) = \{i\} \cup \mathcal{N}_i(t) \quad (6)$$

where choosing i indicates local execution, while choosing any $j \in \mathcal{N}_i(t)$ indicates allocate to neighbor j .

4) *State Transition Probability* ($\mathcal{P}$): The state transition function $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ defines the probability of moving from one global state $s_t \in \mathcal{S}$ to the next state $s_{t+1} \in \mathcal{S}$ given the joint action $\mathbf{a}_t = (a_{1,t}, \dots, a_{N,t}) \in \mathcal{A}$ taken by all agents at time t . Formally,

$$\mathcal{P}(s_{t+1} | s_t, \mathbf{a}_t) = \mathcal{P}(s_{t+1} | s_t, a_{1,t}, \dots, a_{N,t}). \quad (7)$$

5) *Reward Function* ($\mathcal{R}$): All agents share a common reward function R_t designed to maximize the mission completion rate while simultaneously minimizing the overall mission latency. The reward function is defined as:

$$R_t = \omega_{\text{succ}} \cdot N_t^{\text{succ}} - \omega_{\text{lat}} \cdot \sum_{k \in \mathcal{K}_t^{\text{succ}}} L_k - \omega_{\text{fail}} \cdot N_t^{\text{fail}} \quad (8)$$

where N_t^{succ} denotes the total number of missions successfully completed within time step t , $\sum_{k \in \mathcal{K}_t^{\text{succ}}} L_k$ denotes the sum of latencies for all successfully completed missions k at this time step, and N_t^{fail} is the number of failed missions within time step t . The coefficients $\omega_{\text{succ}}, \omega_{\text{lat}}, \omega_{\text{fail}}$ are positive weighting parameters.

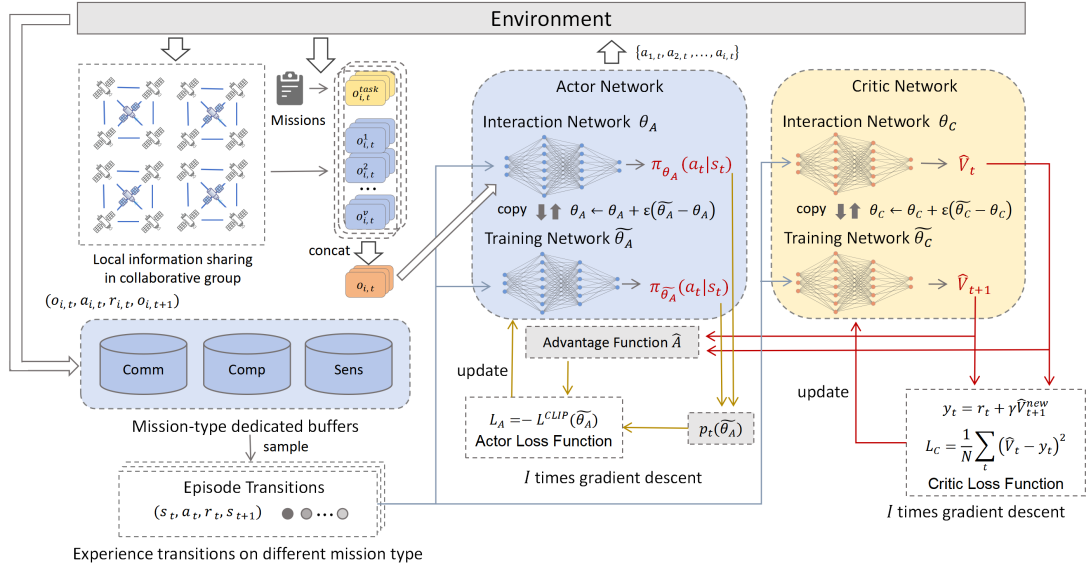


Fig. 2: The architecture of proposed TRMAPPO framework for satellite mission scheduling.

B. TRMAPPO: Transformer-Reptile MAPPO Framework

TRMAPPO integrates two crucial components: Multi-Agent Proximal Policy Optimization (MAPPO) with Transformer-based networks for stable policy learning and capturing complex spatial-temporal dependencies, and a Reptile meta-learning algorithm to effectively address mission heterogeneity and enable rapid adaptation.

To handle intricate inter-agent dynamics characteristic of satellite networks, both the decentralized policy networks and the centralized value network in our TRMAPPO framework are built upon Transformer-based architectures. Specifically, we draw on the model design philosophy of multi-task learning (MTL) to better capture both the commonalities and the specificities across multiple missions. For different mission types τ , the model shares parameters in the lower layers, while assigning an independent output head FC_τ to each mission type. Accordingly, the action output is expressed as

$$\pi_{\theta_i}(a_{i,t} | o_{i,t}) = \text{softmax}(\text{FC}_\tau(\mathbf{h}_{i,t}^{\text{actor}})), \quad (9)$$

where $\mathbf{h}_{i,t}^{\text{actor}}$ is the shared representation derived from the Transformer encoder. Similarly, the value estimation output is expressed by:

$$V_{\psi}(\mathbf{o}_t) = \text{FC}_\tau(\mathbf{h}_t^{\text{critic}}). \quad (10)$$

The actor and critic losses are defined separately as:

$$L^{\text{actor}}(\theta'_\tau) = \mathbb{E}_t [\min(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1 - \epsilon, 1 + \epsilon) \hat{A}_{i,t})], \quad (11)$$

$$L^{\text{critic}}(\psi'_\tau) = \mathbb{E}_t [(V_{\psi'_\tau}(\mathbf{o}_t) - R_{i,t})^2], \quad (12)$$

where $r_{i,t} = \frac{\pi_{\theta'_\tau}(a_{i,t}|o_{i,t})}{\pi_{\theta_{\text{old}}}(a_{i,t}|o_{i,t})}$ and $\hat{A}_{i,t}$ is the advantage estimate for agent i at timestep t . The advantage is computed using Generalized Advantage Estimation (GAE):

$$\hat{A}_{i,t} = \sum_{l=0}^{T-1} (\tau\lambda)^l (R_{i,t+l} + \tau V_{\psi'}(\mathbf{o}_{t+l+1}) - V_{\psi'}(\mathbf{o}_{t+l})), \quad (13)$$

where $R_{i,t}$ is the reward received by agent i at timestep t .

During the inner-loop adaptation phase, for each sampled mission type τ , the global reptile parameters (θ, ψ) are adapted for K gradient steps using the mission-specific experience and the actor-critic losses. For each update step κ , the process is as follows:

$$\theta_{\tau,\kappa+1} = \theta_{\tau,\kappa} - \alpha \nabla_{\theta_{\tau,\kappa}} L^{\text{actor}}(\theta_{\tau,\kappa}), \quad (14)$$

$$\psi_{\tau,\kappa+1} = \psi_{\tau,\kappa} - \alpha \nabla_{\psi_{\tau,\kappa}} L^{\text{critic}}(\psi_{\tau,\kappa}), \quad (15)$$

where $(\theta_{\tau,0}, \psi_{\tau,0}) = (\theta, \psi)$, and α is the inner-loop learning rate. After K steps, the adapted mission-specific parameters become $(\theta'_\tau, \psi'_\tau) = (\theta_{\tau,K}, \psi_{\tau,K})$.

In the outer-loop update, the global reptile parameters are updated by moving them towards the adapted mission-specific parameters:

$$\theta \leftarrow \theta + \beta(\theta'_\tau - \theta), \quad (16)$$

$$\psi \leftarrow \psi + \beta(\psi'_\tau - \psi), \quad (17)$$

where β is the soft update weight in outer-loop training. This update ensures that the global parameters are a good initialization for all missions, allowing for rapid adaptation to new or unseen mission types. To reduce model complexity and training difficulty, a parameter-sharing training scheme is adopted in our model. The comprehensive training procedure for TRMAPPO is presented in Algorithm 1.

IV. SIMULATION RESULTS

A. Simulation Setup

In our experiments, each simulation episode lasts for a duration of 200s, with a time slot interval of $\Delta t = 0.1$ s. We construct a three-layer satellite network architecture with 311 agent satellite nodes, including 174 LEO satellites at 550 km altitude, 100 SSO satellites at 700 km with a 98° inclination, and 37 GEO satellites in geostationary orbits at 35,786 km, jointly providing communication, computation, and sensing

Algorithm 1 TRMAPPO: Transformer-Reptile MAPPO Framework

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1: Initialize: Global actor and critic network with parameters
    $\theta$  and  $\psi$ , mission buffers  $\mathcal{D}_\tau$ , outer update weight  $\beta$ , inner
   steps  $K$ , inner learning rate  $\alpha$ .
2: for each training iteration do
3:   1. Experience Collection
4:   for  $t = 0$  to  $T - 1$  do
5:     Sample mission  $k \sim p(\tau)$ .
6:     Observe joint states  $\mathbf{o}_t$ .
7:     Compute joint action  $\mathbf{a}_t \sim \pi_\theta(\mathbf{o}_t)$ .
8:     Execute  $\mathbf{a}_t$ , observe  $r_t$ ,  $\mathbf{o}_{t+1}$ .
9:     Store  $(\mathbf{o}_t, \mathbf{a}_t, r_t, \mathbf{o}_{t+1})$  in  $\mathcal{D}_\tau$ .
10:  end for
11:  2. TRMAPPO Policy Training
12:  for each mission type  $\tau$  do
13:     $(\theta', \psi') \leftarrow (\theta, \psi)$ 
14:    for  $\kappa = 0$  to  $K - 1$  do
15:      Sample mini-batch  $\mathcal{B}_\tau \sim \mathcal{D}_\tau$ 
16:      Compute GAE  $\hat{A}_{i,t}$  for each agent
17:      Compute actor loss  $\mathcal{L}_\tau^{\text{actor}}(\theta')$  and critic loss
         $\mathcal{L}_\tau^{\text{critic}}(\psi')$ 
18:      Update inner-loop parameters:
19:       $\theta' \leftarrow \theta' - \alpha \nabla_{\theta'} \mathcal{L}_\tau^{\text{actor}}$ 
20:       $\psi' \leftarrow \psi' - \alpha \nabla_{\psi'} \mathcal{L}_\tau^{\text{critic}}$ 
21:    end for
22:    Update actor and critic parameters:
23:     $\theta \leftarrow \theta + \beta(\theta' - \theta)$ 
24:     $\psi \leftarrow \psi + \beta(\psi' - \psi)$ 
25:  end for
26: end for

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services. A inter-satellite link clearance of 150 km is maintained to ensure LoS connectivity. The ISLs are established in the 193.5 THz optical band, with a bandwidth of 1–4 GHz. Each satellite transmits with a power of 0.5–5 W. A four-core CPU computer and a six-camera data acquisition system are deployed on satellites with computing and remote sensing capabilities, respectively.

To capture realistic high-volume service demands, a geographically-aware stochastic mission generation framework is employed, based on real-world demographic and environmental data, including China's population density, GDP density, and tree age distribution. In the training process, the discount factor is set to 0.90, the GAE parameter is 0.98, the policy clipping ratio is 0.2, the inner-loop learning rate is 1×10^{-4} , and the outer-loop step size is set to 1.0.

B. Baseline Algorithms

We compare the proposed TRMAPPO method with five different strategies:

- **Random:** assigns each mission to any available satellite.
- **Greedy:** makes a feasible schedule decision that minimizes the completion time of the current mission.

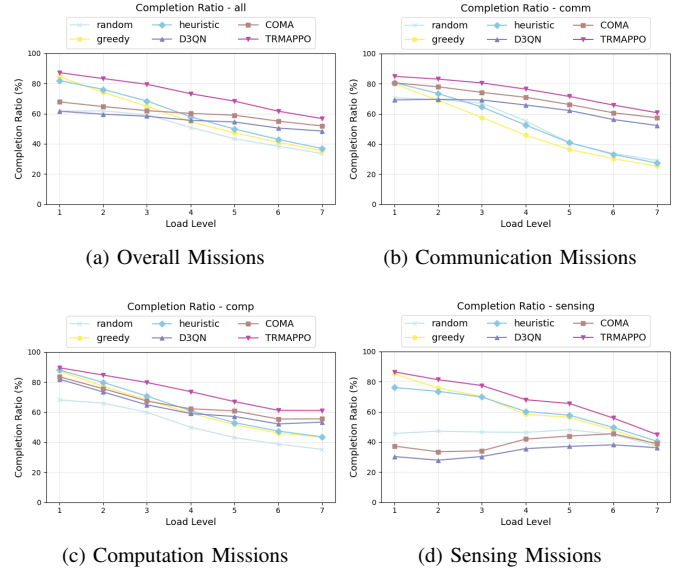


Fig. 3: Mission Completion Ratio of different algorithms with increasing system load levels.

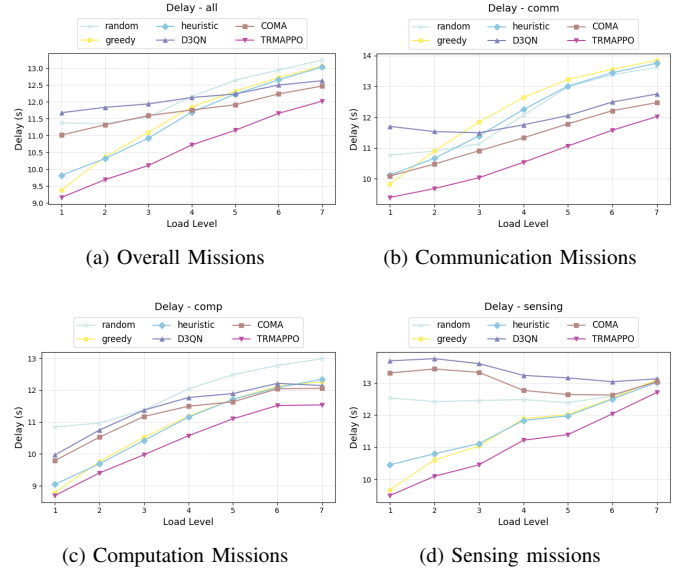


Fig. 4: Average Mission Delay of different algorithms as a function of increasing system load levels.

- **Heuristic:** assigns missions to the satellite with highest weighted score based on available computational capacity, mission queue length, and communication data rate.
- **D3QN** [15]: uses a Deep Q-Network (DDQN) with dueling architecture to reduce the system cost for satellite mission allocation.
- **COMA** [16]: uses an exclusive counterfactual baseline and integrates the CTDE paradigm to achieve efficient and low-complexity satellite mission scheduling.

C. Evaluation Results

The completion ratio and average delay of all algorithms are presented in Fig. 3 and Fig. 4 across various system load levels, respectively. With the increase of mission load, the

completion ratio decreases while the mission delay rises significantly, caused by limited satellite resources. When a large number of missions arrive within a short time window, network congestion and longer processing queues occur, leading to an increased number of failed missions due to deadline violations.

Fig. 3(a) and Fig. 4(a) show that under low to medium load conditions, the Heuristic and Greedy algorithms perform relatively well compared to some DRL baselines and occasionally even surpass them. This is because learning-based policies are trained on high-dimensional average distributions, which can sometimes make them overly cautious in rare scenarios, leading to less effective or even overfitted decisions. When the load level increases, the performance of conventional algorithms deteriorates sharply, due to their myopic decision-making and inability to handle complex resource interdependencies under heavy load, in contrast to learning-based algorithms.

In comparison, under all load conditions, the proposed TRMAPPO consistently achieves the highest completion ratio and the lowest average mission delay. Specifically, it achieves a 13.02% improvement in mission completion ratio and an 8.84% reduction in average delay compared to the strong-performing COMA algorithm under typical medium-load scenarios. Moreover, from Fig. 3(b)–3(d) and Fig. 4(b)–4(d), it can be observed that existing learning-based algorithms struggle to balance the heterogeneous mission characteristics and resource requirements, making it difficult to achieve optimal scheduling. In contrast, the proposed TRMAPPO, empowered by the self-attention mechanism and Reptile meta-learning, effectively captures both environmental and mission features, enabling scheduling decisions that accommodate diverse mission demands, which demonstrates the superior generalization ability and robustness of our proposed strategy.

V. CONCLUSION

This paper addresses the critical challenge of heterogeneous mission scheduling in multi-layer cooperative satellite network. We formulate the heterogeneous mission scheduling problem as a multi-task learning problem, aiming to select appropriate cooperative satellites. We first proposed TRMAPPO, a novel framework that combining Reptile meta-learning with Transformer-based MAPPO networks, which learns a generalized policy adapting to different mission types. It adopts an AC+CTDE architecture with parameter sharing, which significantly reduces the number of parameters and lowers the overall computational complexity. Simulation results show that TRMAPPO consistently outperforms state-of-the-art baselines, achieving up to 13.02% higher mission completion rates and reducing average latency by as much as 8.84%, and exhibits strong generalization under different load conditions. Future work will explore decentralized and lightweight architectures with uncertainty-aware long-horizon planning under practical constraints.

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REFERENCES

- [1] Q. Wang, X. Chen, Q. Qi, M. Li, and W. Gerstacker, "Multiple-Satellite Cooperative Information Communication and Location Sensing in LEO Satellite Constellations," *IEEE Transactions on Wireless Communications*, vol. 24, no. 4, pp. 3346–3361, 2025.
- [2] X. Haoran, Z. Yafeng, and F. Xin, "Resource allocation for emergency communications in distribution network satellite communication system," *China Communications*, vol. 22, no. 2, pp. 95–111, 2025.
- [3] C. D. Bussy-Virat, C. S. Ruf, and A. J. Ridley, "Relationship Between Temporal and Spatial Resolution for a Constellation of GNSS-R Satellites," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 12, no. 1, pp. 16–25, 2019.
- [4] L. Cheng, X. Wang, and X. Sun, "SCODA: Joint Optimization of Task Offloading and Resource Allocation in LEO-Assisted MEC System With Satellite Cooperation," *IEEE Wireless Communications Letters*, vol. 14, no. 10, pp. 3304–3308, 2025.
- [5] C. Yingnan, X. Li, and L. Kai, "Research on Node Selection Algorithm of Near Space Relay Link," in *2022 5th International Conference on Communication Engineering and Technology (ICCET)*, 2022, pp. 27–31.
- [6] W. Zeng, J. Zhang, D. W. K. Ng, B. Ai, and Z. Zhong, "Two-Way Hybrid Terrestrial-Satellite Relaying Systems: Performance Analysis and Relay Selection," *IEEE Transactions on Vehicular Technology*, vol. 68, no. 7, pp. 7011–7023, 2019.
- [7] L. Cheng, G. Feng, Y. Sun, M. Liu, and S. Qin, "Dynamic Computation Offloading in Satellite Edge Computing," in *ICC 2022 - IEEE International Conference on Communications*, 2022, pp. 4721–4726.
- [8] J. Zhou, Q. Yang, L. Zhao, H. Dai, and F. Xiao, "Mobility-Aware Computation Offloading in Satellite Edge Computing Networks," *IEEE Transactions on Mobile Computing*, vol. 23, no. 10, pp. 9135–9149, 2024.
- [9] G. Li, X. Li, J. Li, J. Chen, and X. Shen, "PTMB: An online satellite task scheduling framework based on pre-trained Markov decision process for multi-task scenario," *Knowledge-Based Systems*, vol. 284, p. 111339, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0950705123010870>
- [10] J. Long, Z. Qian, F. Xie, Z. Ding, and L. Liu, "An Improved Multi-Satellite Cooperative Task Planning Method Based on Distributed Multi-agent System," in *2021 13th International Conference on Measuring Technology and Mechatronics Automation (ICMTMA)*, 2021, pp. 539–542.
- [11] G. Sun, J. Wu, Z. Sun, L. He, J. Wang, D. Niyato, A. Jamalipour, and S. Mao, "JC⁵A: Service Delay Minimization for Aerial MEC-Assisted Industrial Cyber-Physical Systems," *IEEE Transactions on Services Computing*, vol. 18, no. 5, pp. 2976–2993, 2025.
- [12] X. Cao, F. Wang, J. Xu, R. Zhang, and S. Cui, "Joint Computation and Communication Cooperation for Energy-Efficient Mobile Edge Computing," *IEEE Internet of Things Journal*, vol. 6, no. 3, pp. 4188–4200, 2019.
- [13] Y. Ding, Q. Zhang, W. Lu, N. Zhao, A. Nallanathan, X. Wang, and X. Yang, "Collaborative Communication and Computation for Secure UAV-Enabled MEC Against Active Aerial Eavesdropping," *IEEE Transactions on Wireless Communications*, vol. 23, no. 11, pp. 15915–15929, 2024.
- [14] Z. Kang, K. Grauman, and F. Sha, "Learning with Whom to Share in Multi-task Feature Learning," *Proceedings of the 28th International Conference on Machine Learning, ICML 2011*, pp. 521–528, 2011.
- [15] J. Wu, M. Jia, N. Zhang, and Q. Guo, "Multi-Agent Deep Reinforcement Learning-Based Computation Offloading in LEO Satellite Edge Computing System," *IEEE Communications Letters*, vol. 28, no. 10, pp. 2352–2356, 2024.
- [16] H. Zhang, H. Zhao, R. Liu, A. Kaushik, X. Gao, and S. Xu, "Collaborative Task Offloading Optimization for Satellite Mobile Edge Computing Using Multi-Agent Deep Reinforcement Learning," *IEEE Transactions on Vehicular Technology*, 2024.